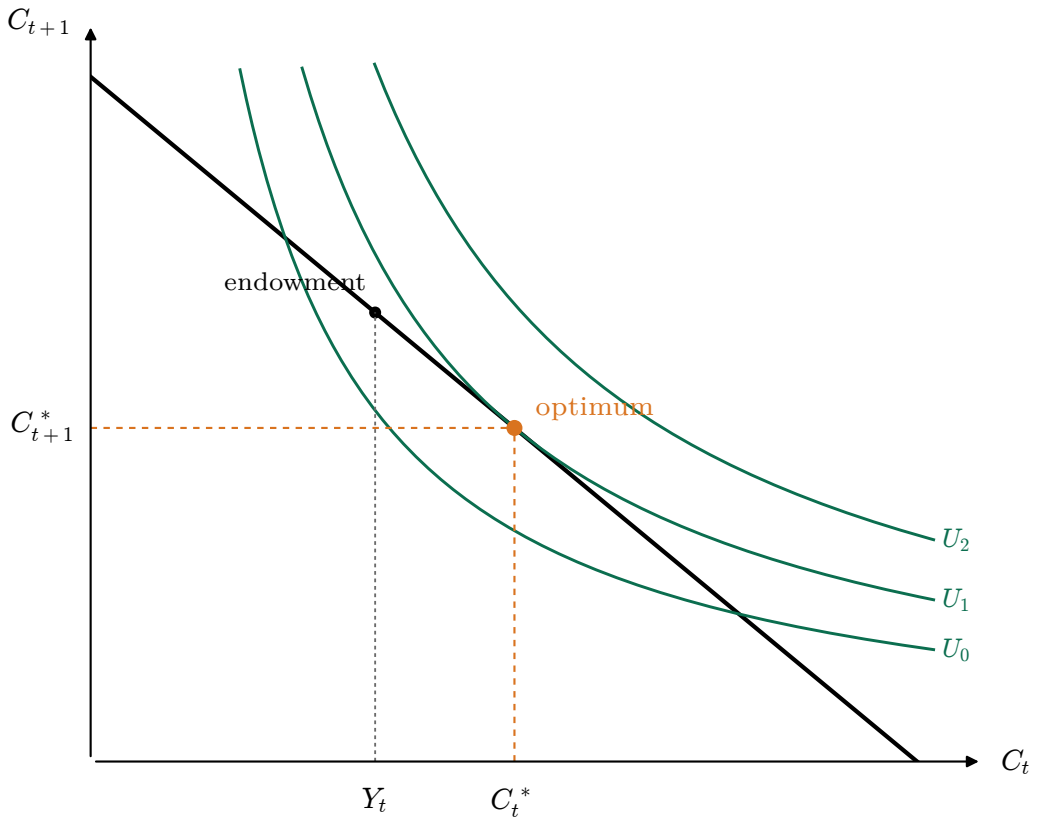

Fig. 2: The optimum is the tangency point: $\dfrac{u'(C_t)}{\beta u'(C_{t+1})} = 1 + r$ (Euler equation). Here $C_t^* > Y_t$: the household borrows.